

Ornstein-Uhlenbeck mapping of First Order Reaction Kinetics at Equilibrium

Abstract

The reaction of α -D-glucose performing mutarotation in tridistilled water to form β -D-glucose at equilibrium is a well-studied and measured S_N1 reaction. This equilibrium reaction is given by $\alpha \rightleftharpoons \beta$, with the literature value₁ of $\alpha \rightarrow \beta$, given by k_{forward} , $4.91 \times 10^{-5} s^{-1}$, and $\beta \rightarrow \alpha$, given by k_{backward} , $2.76 \times 10^{-5} s^{-1}$.

By modelling the mutarotation of α -D-glucose as an OU process, we demonstrate that this stochastic approximation, at macroscopic levels, recovers the classical deterministic rate law as relative fluctuations die down, and converges to the stochastic kinetics of the Gillespie process.

Theory & Code

The OU process is a mean-reverting, Gauss-Markov process characterised by the SDE:

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t \quad (1)$$

To map the mutarotation of α -D-glucose, we must compare the first order rate law of this reaction with the SDE, resulting in this set of parameters:

$$\begin{aligned} dX_t &= d[\alpha] \\ \theta &= k_{\text{forward}} + k_{\text{backward}} \\ \mu &= \frac{k_{\text{backward}}[\alpha_0]}{k_{\text{forward}} + k_{\text{backward}}} \\ \sigma &= \sqrt{2\theta Np(1-p)} \end{aligned}$$

From this set, only θ can be classified as a global parameter given it is the only controlled variable. The other parameters all depend on the initial concentration of α -D-glucose, $[\alpha_0]$.

Instead of building a singular function for the SDE of the OU simulation, it was decided that splitting the SDE into two constituents would be a far cleaner and far easier solution. The mean-reverting term and the noise term was therefore separated into these functions: `mean_revert(N, alpha_count, dt)` and `noise(N, dt)` respectively.

Due to the continuous nature of the SDE, in order to simulate the process numerically, we must discretise it, which was accomplished via the Euler-Maruyama scheme, approximating the continuous process by finite time steps, `dt`, with the mean-reversion term (drift) being scaled by `dt` and the noise term (diffusion) scaled by $\sqrt{dt}$, reflecting the variance of the Wiener increment in the original SDE over a step of length `dt`.

Various situations may arise for individuals who wish to utilise the OU process. From a chemistry perspective, it may be desirable to know whether a certain molecule count has been reached within a time frame. The function `reaction_hit(N, alpha_count, dt, limit, expire)` allows the individual to know whether the reaction has expired of the target molecule count has been reached by implementing conditions.

One of the most crucial comparisons in this project is the difference between the exponential deterministic path and the OU path. The deterministic model gives a noiseless, mean trajectory and when overlayed with the OU process, fluctuations caused by the diffusion term in the SDE are revealed. The standard deviations of the residual data taken across a range of `N` are then fitted on a graph, revealing that although noise appears to have died down, the absolute value of the noise increases along with `N`.

When standard deviation is plotted on log-log axes against `N`, a clearer relationship is observed, with the gradient of the plot being $+1/2$, demonstrating a positive linear relationship. This relationship is seen to be reversed for relative standard deviation, standard deviation over mean, against `N`, with the gradient now

being $-1/2$, validating the initial phenomenon we observe whereby fluctuations appear to die down as N increases.

The second, and arguably the most significant, comparison lies in the Gillespie approach opposed to the OU simulation. It demonstrates clearly de Moivre-Laplace theorem, showing that as N approaches infinity, the normal (Gaussian) can be used to approximate the binomial (Gillespie). As N increases, the Gillespie distribution of α -D-glucose appears smoother, gradually matching the OU distribution. This confirms the convergence of the OU model to the exact stochastic kinetics of the Gillespie process.

Key:

N = number of α -D-glucose molecules

alpha_count = number of initial α -D-glucose molecules

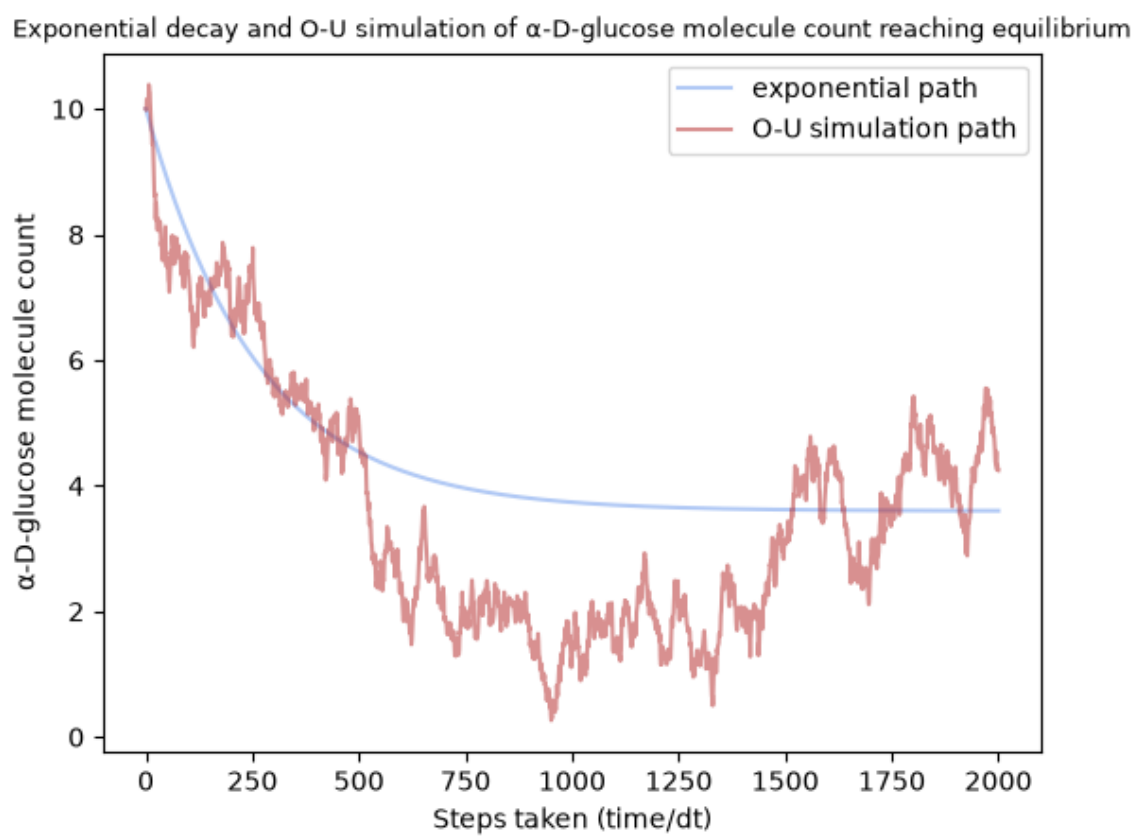
dt = magnitude of one “step”

limit = desired α -D-glucose molecule count

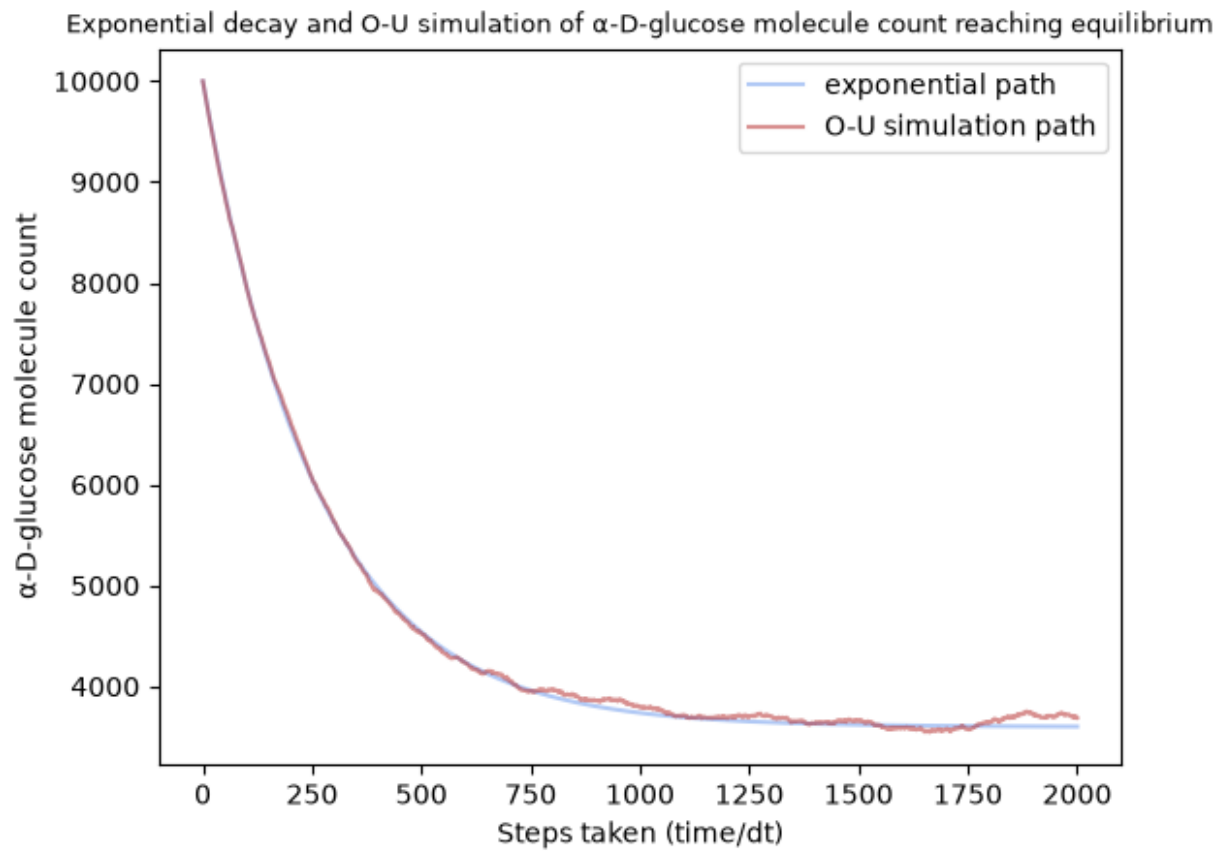
expire = duration of time before reaction expires

Results

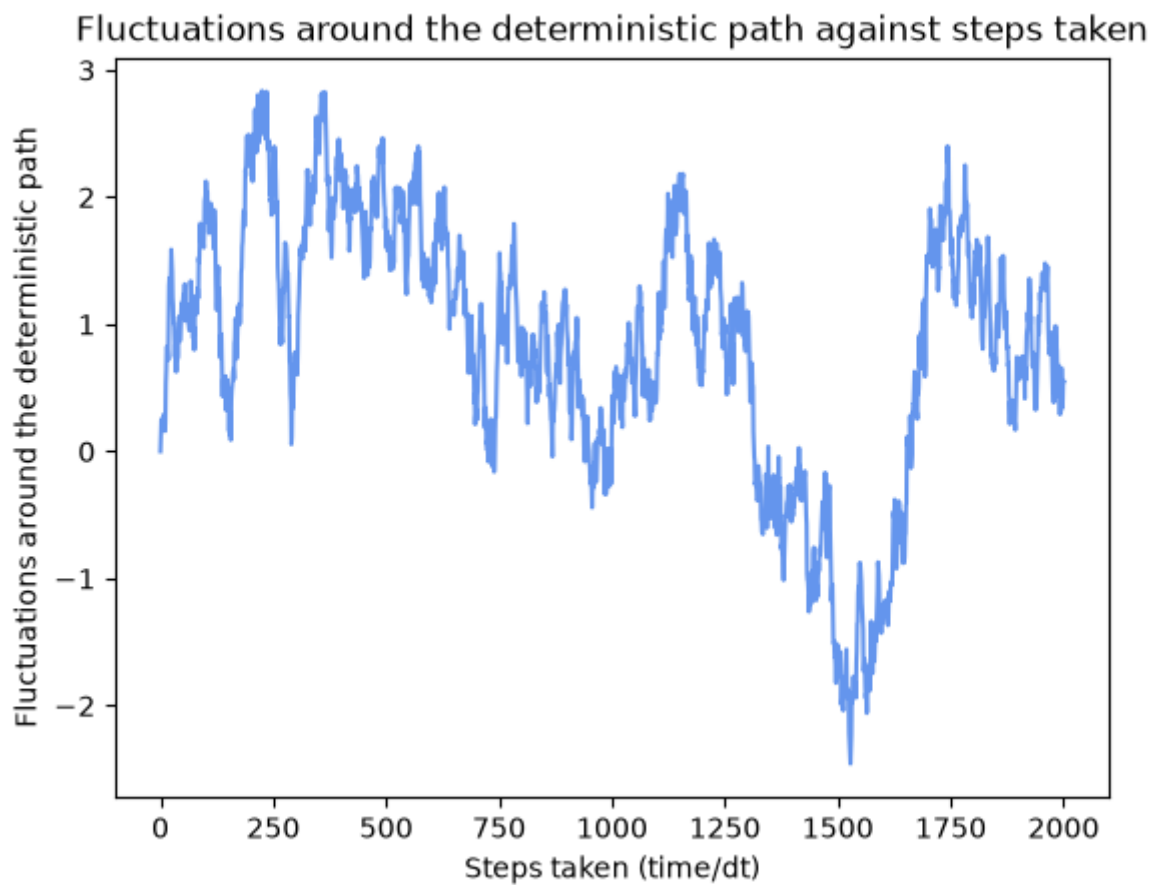
Exponential Decay and O-U simulation of α -D-glucose molecule count reaching eq^m at $N=10$. Fig. 1



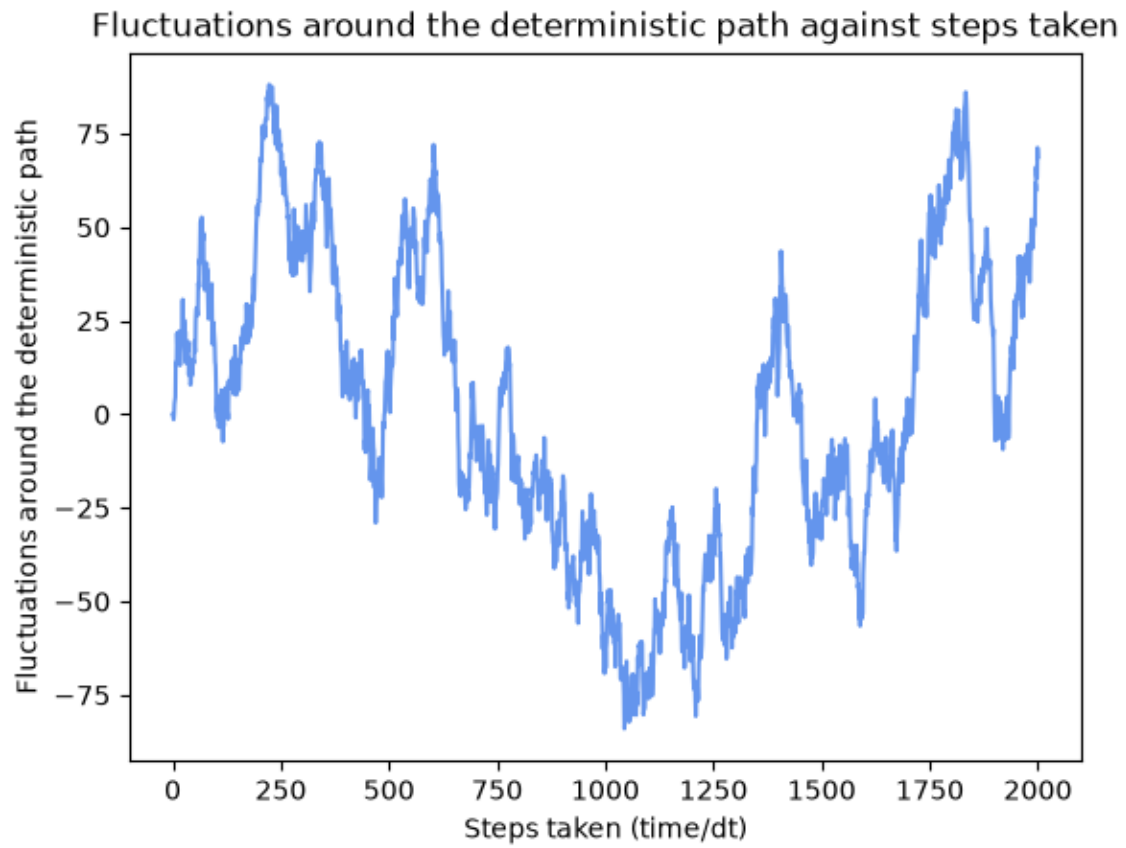
Exponential Decay and O-U simulation of α -D-glucose molecule count reaching eq^m at $N=10000$. Fig. 2



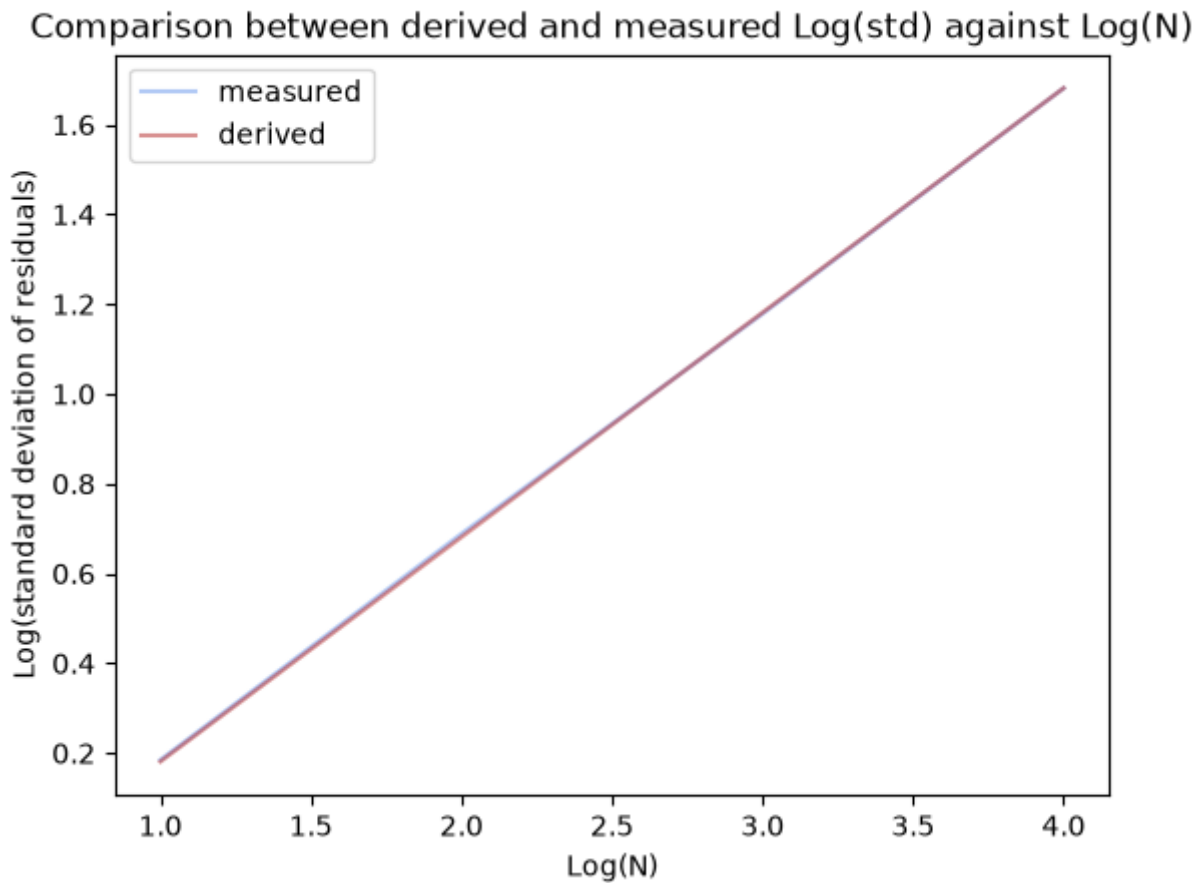
Fluctuations around the deterministic path against steps taken at $N=10$. Fig. 3



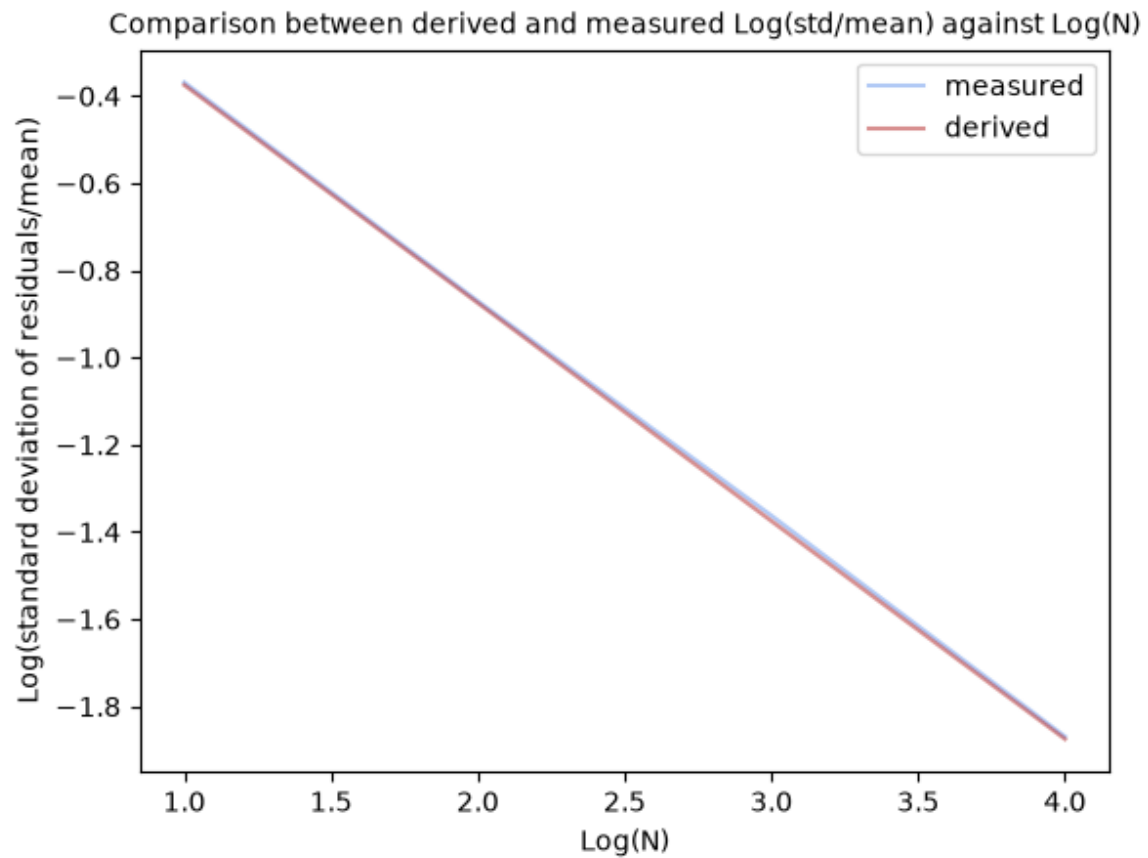
Fluctuations around the deterministic path against steps taken at $N=10000$. Fig. 4



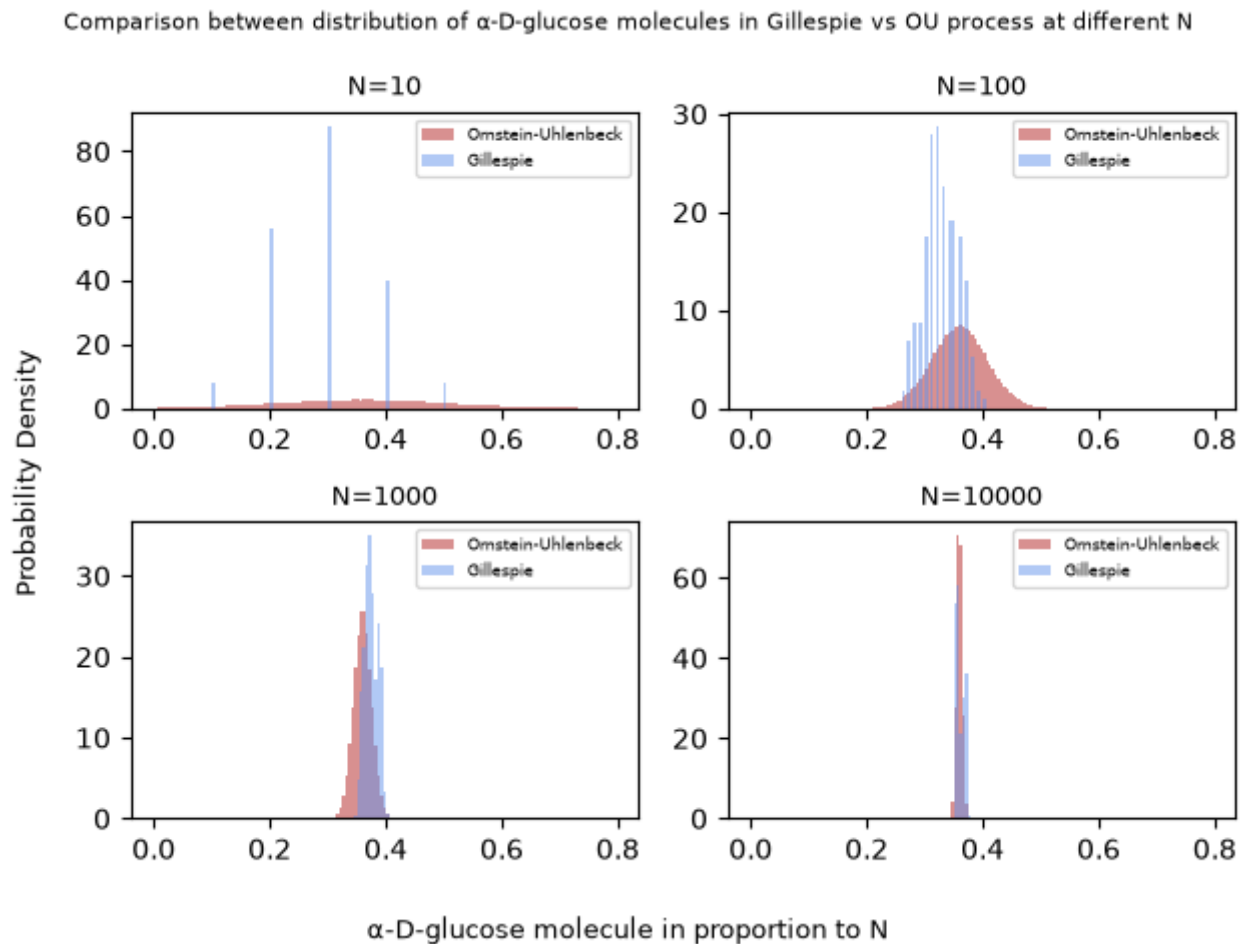
Comparison between derived and measured $\text{Log}_{10}(\text{std}_{\text{absolute}})$ against $\text{Log}_{10}(N)$ Fig. 5



Comparison between derived and measured $\text{Log}_{10}(\text{std}_{\text{relative}})$ against $\text{Log}_{10}(N)$ Fig. 6



Comparisons between distributions of α -D-glucose molecules in Gillespie vs OU at different N . Fig. 7



Analysis

As established before, our hypothesis is that as N increases, the noise surrounding the path relatively decreases, but the absolute noise, or rather standard deviation from the path increases. This is demonstrated via comparisons between *Fig. 1* and *Fig. 2*, and comparisons between *Fig. 3* and *Fig. 4*. Of the former, we see the fluctuations diminishing from $N=10$ to $N=10000$, but upon closer inspection of the residuals, we see that the values of the fluctuations are much larger at $N=10000$ than at $N=10$. The relationship between standard deviation (noise) and N , number of molecules, is demonstrated as such:

$$std_{absolute} = \sqrt{Np(1-p)} \quad (2)$$

$$std_{relative} = \frac{\sqrt{Np(1-p)}}{\mu} \quad (3)$$

Equation (2) can be rewritten to give:

$$std_{relative} = \frac{\sqrt{1-p}}{\sqrt{N}\sqrt{p}} \quad (4)$$

From these equations, the relationship between standard deviation and N becomes clearer to us.

If we take the $\log_{10}$ of both sides for equations (1) and (3), we yield:

$$\log_{10}(std_{absolute}) = \frac{1}{2}\log_{10}(N) + \frac{1}{2}\log_{10}(p(1-p)) \quad (5)$$

$$\log_{10}(std_{relative}) = -\frac{1}{2}\log_{10}(N) + \frac{1}{2}[\log_{10}(1-p) - \log_{10}(p)] \quad (6)$$

By comparing equations (5) and (6) with $y = mx + c$, we obtain *Fig. 5* and *Fig. 6*, where y corresponds to $\log_{10}(std_{absolute})$ and $\log_{10}(std_{relative})$, m corresponds to $\frac{1}{2}$ and $-\frac{1}{2}$, x corresponds to $\log_{10}(N)$ in both cases, and c corresponds to $\frac{1}{2}\log_{10}(p(1-p))$ and $\frac{1}{2}[\log_{10}(1-p) - \log_{10}(p)]$.

This phenomenon is further validated by the convergence of the distribution of α -D-glucose molecules towards the mean value Np , or more accurately in *Fig. 7*, the distribution concentrates towards the fractional concentration of α -D-glucose molecules at equilibrium.

Author's Notes

The Ornstein-Uhlenbeck process is an amazing model for simulating mean reversion in the finance world, being applicable to many areas of the market such as interest rates, FX markets, commodities etc. Through this project, my understanding of the OU process has increased considerably, allowing me to view the financial market in a different angle.

A continuation of the content learnt throughout this project would be utilising the Vasicek model with historical data regarding interest rates. I believe this will be another opportunity for a new perspective to be learnt.

References

1. Lin, C.E., Yu, C.J., Chen, C.L., Chou, L.D., & Chou, C. (2010). Kinetics of Glucose Mutarotation Assessed by an Equal-Amplitude Paired Polarized Heterodyne Polarimeter. *The Journal of Physical Chemistry A*, 114(4), 1665-1669.